No Universal Temperature: Quality Falls at Model-Specific Cliffs Across Tasks, Sizes, and Families

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Abstract

Softmax temperature is the most widely exposed decoding setting, and practitioners choose it by folklore: 0.2 for code, 0.7 for chat, above 1 for creativity. We ask how output diversity and quality change with temperature on open-ended, factual, and coding tasks, and whether the useful range depends on model size and family. We sample 10 open-weight models from four families at 11 temperatures from 0 to 2.0 on stories, closed-book trivia, and executed code, for about 277,000 samples including two control runs. Quality does not decline gradually. It stays nearly flat and then falls at a point specific to each model: story coherence drops below 90% of peak at $T \approx 0.84$ for Qwen2.5-0.5B and $T \approx 1.31$ for Qwen2.5-14B. Code does not collapse just above 0.8 as folklore suggests: no model drops significantly from 0.6 to 0.8, and models of 3B and larger in the Qwen family, plus Gemma, stay within 90% of peak to $T \approx 1.1$–1.4. Factual accuracy loses only 0.7–5.2 points at T of 0.8–1.0, with no further coverage gain. Larger models tolerate higher temperature on all three tasks in the Qwen family, but only on QA for Llama, and Llama curves differ from the others by 4–7 times the seed noise floor. Temperatures below about 0.8 are safe on nearly every model tested but do little for diversity; above 1.0, the right setting has to be checked per model.

1 Introduction

Temperature is the one decoding setting that nearly every large language model (LLM) interface exposes, and most users set it by habit. The common advice says to use a low value for code, about 0.7 for chat, and a value above 1 for creative work. Our main question is whether this advice matches what temperature actually does.

The answer matters because temperature trades off two things people want. Higher values make outputs more varied, which helps brainstorming, creative writing, and any workflow that samples several candidates. They also make outputs more likely to be wrong or broken. If diversity stops improving while quality keeps falling, a high setting buys nothing. If the safe range depends on the model, a single default is wrong for most deployments.

Existing studies answer parts of this question. Renze and Guven [2024] found no significant effect of temperature between 0 and 1 on multiple-choice problem solving. Peeperkorn et al. [2024] found only a weak link between temperature and story novelty for one fixed model. Chen et al. [2021] tuned temperature for code on one model family. However, each of these studies looks at a single kind of task, so none shows whether a limit found on one task carries over to another or how it moves with model size and family. On the other hand, the folklore makes exactly such claims. We state one version as a testable hypothesis: open-ended diversity plateaus above $T = 1.2$ while coherence drops 20–40%; factual accuracy degrades above 0.7 with no useful diversity gain; code has a narrow useful range of 0.2–0.6 with a sharp collapse above 0.8; smaller models have narrower useful ranges; and curve shapes differ across families at the same scale.

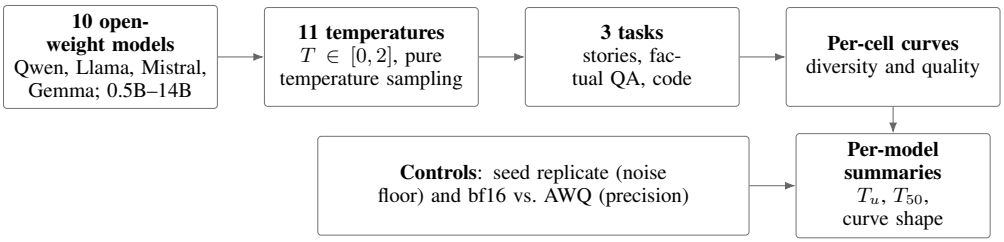


Figure 1: Study design. We cross 10 models, 11 temperatures, and 3 tasks, measure diversity and quality in every cell, and summarize each model by where its quality falls. Two control runs set the noise floors for comparing curves.

We test this hypothesis with one controlled grid (figure 1). We run 10 open-weight models, two size ladders plus two models from other families, at 11 temperatures from 0 to 2.0 on stories, closed-book trivia, and executed code. All models use pure temperature sampling, so no other decoding setting hides the effect. For each cell we measure both diversity and quality, then derive each model's useful temperature range.

We find that temperature fails abruptly and at a model-specific point. Story coherence falls below 90% of peak at $T \approx 0.84$ for Qwen2.5-0.5B and at $T \approx 1.31$ for Qwen2.5-14B. No fixed threshold describes all models. The code clause is largely wrong: no model drops significantly from $T = 0.6$ to 0.8, and the largest relative drop from 0.6 to 1.0 is 28%. Greedy decoding is best or tied for best on pass@1 for 5 of 10 models, while sampling four times at $T = 0.8$ adds 6–18 points over a single greedy attempt. Factual accuracy does fall above 0.7, but by only 0.7–5.2 points at T of 0.8–1.0.

Our contributions are:

- We measure diversity and quality on one grid across three tasks, 10 models, and temperatures up to 2.0, with seed and precision controls.
- We test five common claims about temperature and give each a clause-level verdict: one partly supported, one mostly supported with a small effect, one refuted as stated, and two supported with qualifications.
- We show that the useful range depends on both size and family, and that Llama models break earlier on stories than same-size models from other families.
- We show that much of the apparent diversity plateau is a measurement artifact: distinct-2 hits its ceiling and embedding diversity becomes unreliable on degenerate text.

Section 2 reviews prior work, section 3 describes our design, section 4 presents results for each claim, section 5 covers interpretation and limitations, and section 6 concludes.

2 Related Work

This project had no systematic literature search, so we position our work against four directly relevant studies rather than claim to cover the field.

Temperature and degenerate text. Holtzman et al. [2020] showed that high-randomness sampling produces incoherent text and proposed nucleus sampling to fix it. We do not compare samplers. Instead, we hold every other setting off and ask where pure temperature sampling breaks for each model.

Temperature and accuracy. Renze and Guven [2024] tested nine LLMs on multiple-choice problems at temperatures from 0 to 1.6 and found no significant effect between 0 and 1. Our tasks use open-ended trivia and code rather than multiple choice. With paired tests, we detect a small but significant accuracy loss at T of 0.8–1.0 for 9 of 10 models, which agrees with theirs on the small size of the effect but not on its absence.

Temperature and creativity. Peeperkorn et al. [2024] asked whether temperature is the "creativity parameter" of LLMs, using one fixed context, model, and prompt. They found that temperature is weakly correlated with novelty and moderately correlated with incoherence. Building on this,

Model	Checkpoint	Params	Precision
Qwen2.5-0.5B, -1.5B	`Qwen/Qwen2.5-{0.5,1.5}B-Instruct`	0.5B, 1.5B	bf16
Qwen2.5-3B, -7B, -14B	`Qwen/Qwen2.5-{3,7,14}B-Instruct-AWQ`	3.1B, 7.6B, 14.8B	AWQ-INT4
Llama-3.2-1B	`unsloth/Llama-3.2-1B-Instruct`	1.2B	bf16
Llama-3.2-3B	`AMead10/Llama-3.2-3B-Instruct-AWQ`	3.2B	AWQ-INT4
Llama-3.1-8B	`hugging-quants/Meta-Llama-3.1-8B-Instruct-AWQ-INT4`	8.0B	AWQ-INT4
Mistral-7B	`solidrust/Mistral-7B-Instruct-v0.3-AWQ`	7.2B	AWQ-INT4
Gemma-2-9B	`hugging-quants/gemma-2-9b-it-AWQ-INT4`	9.2B	AWQ-INT4
Control: seed replicate	Qwen2.5-0.5B, different sampling seeds	0.5B	bf16
Control: precision	`Qwen/Qwen2.5-7B-Instruct`	7.6B	bf16

Table 1: Subject models. Each family is represented by one checkpoint per size, and the Llama ladder spans versions 3.2 and 3.1.

we track diversity and coherence together across 10 models, which shows how far the point of incoherence moves between models.

Temperature and code. Chen et al. [2021] reported that for a 679M-parameter Codex model the best temperature was 0.2 for pass@1 and 0.8 for pass@100, because higher temperatures give more diverse samples. Our pass@4 results follow the same pattern over a wider range of models. Unlike that study, we put code on the same grid as other tasks and test whether a narrow useful range holds across sizes and families.

3 Methodology

Problem formulation. Given logits z_i for the next token, temperature sampling draws token i with probability

$$p_i(T) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}, \tag{1}$$

and $T = 0$ denotes greedy decoding. For a model m and a quality curve $q_m(T)$, we define the useful range as the temperatures that keep quality near the model's own peak:

$$\mathcal{U}_m = \{\, T : q_m(T) \geq 0.9 \max_{T'} q_m(T') \,\}, \tag{2}$$

with linear interpolation between grid points. T_{lo} and T_u are its lower and upper bounds, and T_{50} is where quality falls to 50% of peak. We choose a relative threshold because models differ widely in absolute quality. This choice means that T_u says how far a model can go without losing much of what it has, not how good it is. For example, Qwen2.5-0.5B peaks at a coherent rate of only 0.85.

Hypotheses. We split the claim into five sub-hypotheses. **H1 (open-ended):** diversity gain is much smaller above $T = 1.2$, while coherence at $T \geq 1.4$ is 20–40% below peak. **H2 (factual QA):** accuracy above 0.7 falls, with little gain in coverage. **H3 (code):** the useful range is about 0.2–0.6, with a sharp collapse above 0.8. **H4 (size):** T_u grows with model size. **H5 (family):** curves differ across families at matched size.

Models. Table 1 lists the models. No API key was available, so all subjects are open-weight checkpoints that we run locally with vLLM 0.32 on shared NVIDIA RTX A6000 GPUs. We did not test GPT, Claude, or Gemini. All runs use pure temperature sampling: `top_p=1`, `top_k` off, `min_p=0`, `repetition_penalty=1`, and `generation_config="vllm"` so that checkpoint defaults do not apply.

Design. We use temperatures $\{0, 0.2, 0.4, 0.6, 0.7, 0.8, 1.0, 1.2, 1.4, 1.6, 2.0\}$, with 4 samples per (prompt, T) and 1 at the deterministic $T = 0$. This gives 23,124 samples per run and about 277,000 across 12 runs, and all runs completed all 6,204 (prompt, T) cells.

Datasets. We use three tasks, each with one prompt template. **Stories:** 100 WRITINGPROMPTS [Fan et al., 2018] test prompts, up to 256 tokens. **Factual QA:** 300 closed-book TRIVIAQA [Joshi et al., 2017] validation questions, up to 24 tokens. **Code:** all 164 HUMANEVAL [Chen et al., 2021] problems, up to 448 tokens, run against the official tests in a resource-limited subprocess.

Metrics. For stories, diversity is the mean pairwise cosine distance of `all-mpnet-base-v2` embeddings across samples for a prompt (semantic diversity), plus distinct-2 over pooled samples.

Claim	Verdict	Evidence
H1: open-ended diversity plateaus above 1.2, coherence drops 20–40%	Partly supported	Measured diversity stops rising in 9/10 models but peaks at T = 1.0–1.4 and then falls. Judge coherence falls 48–62% at $T = 1.4$ in 9/10 models.
H2: above 0.7, QA accuracy degrades with no diversity gain	Mostly supported, small effect	Accuracy at T = 0.8–1.0 is lower in all 10 models, significantly in 9, by at most 5.2 points. coverage@4 stops improving above 0.7.
H3: code has useful range 0.2–0.6, collapse above 0.8	Refuted as stated	No significant drop from 0.6 to 0.8. T_{lo} = 0 everywhere. Four models reach T_u of 1.07–1.42.
H4: smaller models have narrower useful ranges	Supported for Qwen, weak for Llama	Qwen: all three tasks. Llama: QA only.
H5: curve shape differs across families	Supported	Llama gaps are 4–7 times the seed floor. Qwen and Mistral are close to each other.

Table 2: Verdict on each clause of the hypothesis.

Quality comes from a Qwen3-8B judge that scores coherence from 1 to 9 as a probability-weighted digit, on 2 of 4 samples. We call the share of stories scoring at least 5 the coherent rate, and we also track lexical validity. All 25,200 judgments passed our logprob validity checks. For QA, we measure per-sample accuracy by normalized alias match, coverage@4 (any of 4 samples correct), and distinct answers. For code, we measure pass@1, pass@4, syntax validity, and distinct programs by AST. Quality for T_u is the coherent rate for stories, accuracy for QA, and pass@1 for code. Greedy decoding and the conventional $T = 0.7$ and $T = 1.0$ settings serve as baselines, and each model is compared with its own peak.

Statistics. We use an item-level bootstrap with 2,000 resamples for 95% confidence intervals (CIs) and paired sign-flip permutation tests with 10,000 permutations and Holm correction [Holm, 1979]. Holm correction runs over all 12 runs including controls. For size trends, we report Spearman correlations and bootstrap contrasts between the smallest and largest model of each ladder. The plateau test is the ratio of the diversity slope over $[1.2, 2.0]$ to the slope over $[0.2, 1.2]$. For family comparisons, the shape gap is the mean absolute difference between two curves, each normalized by its own maximum. We compare it with two measured floors: a seed replicate and a bf16-versus-AWQ control.

Changes from the plan. Several analysis choices came after seeing partial data, and we report them so readers can weigh the results. Shared-GPU memory moved the 14B and 3B models to AWQ, so precision co-varies with size in both ladders (bf16 at 1.5B and below, AWQ at 3B and above). We chose the coherent rate over the raw judge score for story T_u after seeing Qwen2.5-0.5B and Llama-1B results, because the judge rarely scores below 3. We edited the judge rubric once after inspecting five samples, before any scored run. We added the seed replicate, Holm correction, and ladder contrasts after the initial plan.

4 Results

Table 2 summarizes our verdicts, and figure 2 shows the main pattern: quality holds near each model's peak and then falls, at a point that differs by model and task. Figure 3 shows all raw curves. Quality falls most abruptly on stories and more gradually on QA.

4.1 H1: open-ended generation

Does diversity plateau above 1.2? In a sense it does, but "plateau" is the wrong shape. Semantic diversity rises, peaks at T of 1.0–1.4, and then falls in 9 of 10 models (table 3). Qwen2.5-14B is the exception, with no slowdown between the two ranges (ratio 0.98, 95% CI 0.69–1.30).

Most of this pattern comes from the measures, not the models. Distinct-2 flattens because it reaches its ceiling of 1.0. We read the fall in semantic diversity as an embedding artifact: degenerate text appears to embed similarly whatever it says, so a falling score does not mean outputs are less varied. We did not test this reading directly. Restricting to prompts where both judged samples are coherent supports it, because semantic diversity then rises at almost every temperature until too few coherent

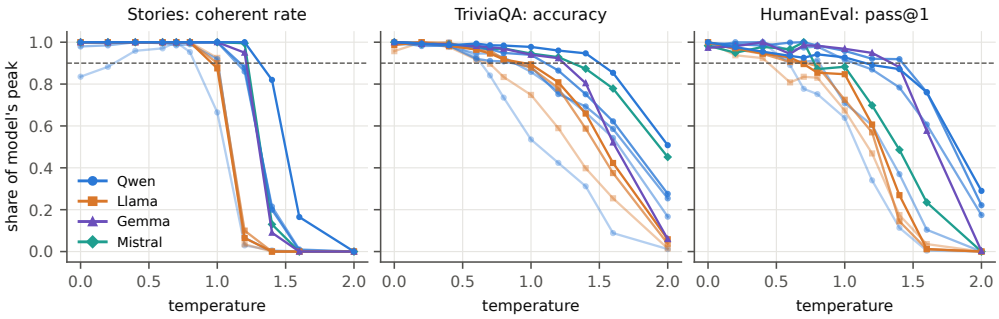


Figure 2: Quality relative to each model's own peak, with darker lines for larger models in a family. The dashed line marks 90%, which defines T_u. On stories, most models go from near their peak to near zero within one or two grid steps, and the break ranges from $T \approx 0.8$ to 1.3 across models.

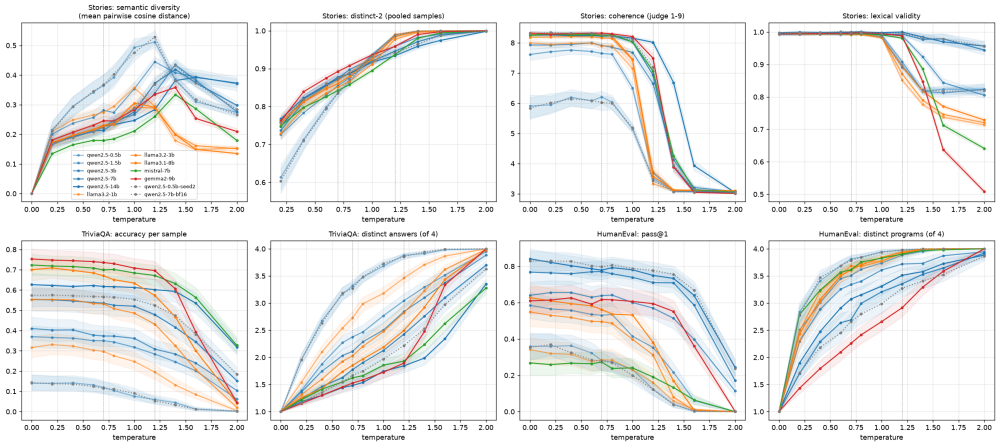


Figure 3: All metrics by temperature for all models and controls, with bands showing bootstrap 95% CIs. Vertical dashed lines mark $T = 0.7$ and 1.2. Dotted gray lines are the control runs, which closely follow the matching main runs in shape.

groups remain. For example, Gemma rises from 0.18 at $T = 0.2$ to 0.37 at $T = 1.4$. This is a selected subset, but it suggests that coherence loss, not saturation, ends useful diversity (figure 4).

Does coherence drop 20–40%? This prediction is met only in passing. At $T = 1.4$ Qwen2.5-14B falls 19%, close to the range, while the other nine models lose 48–62%. These figures understate the damage: the judge rarely scores below 3, so the raw scale cannot fall more than about 64%. The coherent rate shows it better, falling from about 1.0 to about 0 within one or two grid steps. For example, Mistral goes from 0.995 at $T = 1.2$ to 0.13 at $T = 1.4$.

4.2 H2: factual QA

Does accuracy degrade above 0.7? Yes, but the loss is small (table 4). Accuracy at T of 0.8–1.0 is below that at $T \leq 0.7$ in all 10 models and significantly so in 9, by 2.0–5.2 points. Qwen2.5-14B loses 0.7 points, which is not significant ($p = .217$). The decline stays gradual up to about $T = 1.2$ and steepens beyond it.

Does the extra diversity help? Below 0.7 it does: coverage@4 at $T = 0.7$ is 2.7–9.7 points above greedy accuracy. Above 0.7 it does not. Moving from 0.7 to 1.0 changes coverage by -5.0 to +1.7 points, -0.8 on average, even though distinct answers keep rising. The extra diversity is mostly more wrong answers.

Model	Peak diversity T	Slope ratio		Judge drop from peak			Coherent rate	
		Semantic	Distinct-2	T=1.2	T=1.4	T=2.0	T=1.2	T=1.4
Qwen2.5-0.5B	1.2	−1.00	0.03	44%	50%	50%	0.03	0.01
Qwen2.5-1.5B	1.2	−0.38	0.07	53%	60%	61%	0.07	0.00
Qwen2.5-3B	1.4	−0.45	0.23	17%	48%	62%	0.86	0.22
Qwen2.5-7B	1.4	−0.47	0.38	17%	53%	64%	0.88	0.20
Qwen2.5-14B	**1.6**	**0.98**	**0.43**	**3%**	**19%**	63%	**1.00**	**0.82**
Llama-3.2-1B	1.0	−2.09	0.05	58%	61%	62%	0.04	0.00
Llama-3.2-3B	1.0	−1.41	0.12	55%	62%	62%	0.10	0.00
Llama-3.1-8B	1.0	−1.65	0.09	57%	62%	63%	0.07	0.00
Mistral-7B	1.4	−0.81	0.42	15%	49%	63%	0.995	0.13
Gemma-2-9B	1.4	−1.02	0.26	10%	53%	64%	0.95	0.09

Table 3: Open-ended generation. The slope ratio divides the diversity slope over $T \in [1.2, 2.0]$ by the slope over $[0.2, 1.2]$. Most models lose about half their judge score by $T = 1.4$, and the coherent rate shows the loss is near total. Best value in each column is in **bold**, except at $T = 2.0$, where all models sit at the judge's floor.

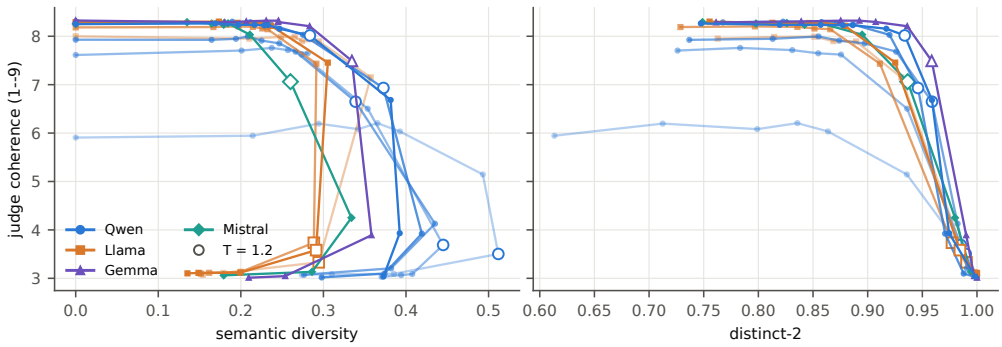


Figure 4: Diversity against coherence as T rises, with hollow markers at $T = 1.2$. Coherence stays flat while diversity rises, then drops steeply. *(Left)*: semantic diversity turns back on itself past the cliff. *(Right)*: distinct-2 presses against its ceiling.

4.3 H3: code generation

Is there a sharp collapse just above 0.8? No (table 5). No model drops significantly from 0.6 to 0.8 after Holm correction. From 0.6 to 1.0 four models drop significantly, all at 3B or below, and the largest relative drop is 28% (Qwen2.5-0.5B), under the 30% we set as our example of a sharp drop in the plan.

Is the useful range 0.2–0.6? The lower bound is not supported: $T_{\text{lo}} = 0$ for every model, and five models reach their best pass@1 at greedy. The upper bound varies from 0.44 to 1.42, and Qwen2.5-3B, -7B, -14B and Gemma all stay within 90% of peak to at least 1.07.

Where does temperature pay off? It pays off when several attempts can be tested. At $T = 0.8$, pass@4 exceeds greedy pass@1 by 6–18 points in every model, led by Qwen2.5-3B at +18.3. Across models, the best temperature for pass@4 ranges from 0.4 to 1.4, and for five models it is 0.7 or 0.8.

Where is the real collapse? It comes later and is syntactic (table 8). Pooled across models, syntax errors go from 2% at $T = 0.8$ to 17% at 1.2, 45% at 1.4, and 93% at 2.0.

4.4 H4: model size

Larger models tolerate higher temperature in the Qwen family on all three tasks, though the code contrast only just clears zero and is not monotonic, since Qwen2.5-14B has a lower T_u than Qwen2.5-7B (table 6, figure 5). In the Llama family the trend holds on QA only. Story T_u is 1.01 at 1B, 1.00 at 3B, and 0.96 at 8B. Code T_{50}, which is less sensitive than T_u, also rises with

	Mean accuracy						coverage@4	
Model	Greedy	$T \leq 0.7$	T 0.8–1.0	**Diff. (pt)**	**Holm** p	$T \geq 1.2$	T=0.7	T=1.0
Qwen2.5-0.5B	.140	.134	.090	−4.4	<.001	.030	.217	.167
Qwen2.5-1.5B	.370	.360	.330	−2.9	.001	.198	.467	.430
Qwen2.5-3B	.410	.393	.367	−2.6	.001	.234	.447	.463
Qwen2.5-7B	.553	.545	.522	−2.2	.001	.348	.593	.603
Qwen2.5-14B	.627	.621	.615	**−0.7**	.217	.512	.653	.660
Llama-3.2-1B	.317	.314	.262	−5.2	<.001	.104	.410	.383
Llama-3.2-3B	.553	.545	.497	−4.8	<.001	.246	.613	.623
Llama-3.1-8B	.700	.692	.642	−5.0	<.001	.346	.763	.760
Mistral-7B	.723	.713	.693	−2.0	<.001	**.548**	.753	.757
Gemma-2-9B	**.753**	**.744**	**.719**	−2.5	.001	.435	**.810**	**.800**

Table 4: Factual QA. The difference compares T of 0.8–1.0 with $T \leq 0.7$. Accuracy falls in every model and significantly in nine, but only by a few points. coverage@4 shows no consistent gain from 0.7 to 1.0. p-values use 10,000 permutations, so "<.001" is near the resolution limit. Best values are in **bold**.

					pass@1					
Model	**Greedy**	**Best** T	T_u **(95% CI)**	T_{50}	0.6	0.8	1.0	1.2	**pass@4 at 0.8**	**Holm** p**, 0.6 vs. 1.0**
Qwen2.5-0.5B	.360	0.4	0.58 (0.41–0.64)	1.09	.323	.273	.232	.123	.476	<.001
Qwen2.5-1.5B	.585	0	0.81 (0.34–0.87)	1.29	.535	.535	.415	.354	.750	<.001
Qwen2.5-3B	.640	0.2	1.07 (0.81–1.22)	1.70	.630	.642	.601	.570	.823	.916
Qwen2.5-7B	.768	0.7	**1.42** (1.07–1.46)	1.79	.770	.759	.739	.710	.860	.392
Qwen2.5-14B	**.841**	0	1.15 (0.56–1.43)	**1.82**	**.787**	**.793**	**.780**	**.750**	**.902**	.946
Llama-3.2-1B	.341	0	0.44 (0.16–0.68)	1.17	.276	.284	.230	.160	.421	.038
Llama-3.2-3B	.549	0	0.72 (0.31–0.88)	1.23	.497	.486	.398	.312	.707	<.001
Llama-3.1-8B	.628	0	0.69 (0.37–0.91)	1.26	.582	.537	.532	.381	.744	.085
Mistral-7B	.268	0.7	0.78 (0.14–1.05)	1.39	.264	.238	.241	.191	.421	.946
Gemma-2-9B	.610	0.4	1.34 (0.60–1.41)	1.65	.591	.616	.607	.595	.707	.946

Table 5: Code generation. Best T maximizes pass@1. Larger models hold their pass@1 far past $T = 0.8$, while the four significant drops from 0.6 to 1.0 are all at 3B or below. T_u CIs are wide because curves are flat near the threshold. Best values are in **bold**.

size: 1.09 to 1.82 for Qwen and 1.17 to 1.26 for Llama. Spearman values within a ladder rest on 3–5 points, so we rely on the bootstrap contrasts instead.

4.5 H5: family differences at matched scale

Curve shapes differ across families, but not evenly (table 7, figure 6). Llama differs from the other three families by 4–7 times the seed floor, and it peaks at $T = 1.0$ rather than 1.4. The 1–1.5B and 3B Qwen-versus-Llama pairs show the same result, with gaps of 0.225 and 0.224. Gemma sits between the groups, and Qwen and Mistral are close to each other.

On stories, family matters more than size in this sample. Llama-8B breaks at $T_u = 0.96$, close to Llama-1B at 1.01, while the other 7–9B models hold to 1.17–1.22. The evidence is weaker on the other two tasks: the seed floors for QA accuracy (0.061) and pass@1 (0.051) are larger, and the largest 7–9B gaps reach only 0.115 and 0.213.

4.6 Controls and failure modes

Seed. Two sampling-seed runs of Qwen2.5-0.5B differ at no temperature on any metric.

Precision. Comparing Qwen2.5-7B in bf16 and AWQ, curve shapes sit at the seed floor (semantic diversity gap 0.026). Levels do differ: greedy pass@1 is .829 versus .768, and greedy QA accuracy is .573 versus .553. AWQ therefore does not explain the shape differences above, but absolute levels should not be compared across precisions.

Task	Qwen 0.5B→14B ΔT_u	Llama 1B→8B ΔT_u	Spearman ρ, 10 models (p)
Stories	+0.47 (0.43, 0.55)	−0.05 (−0.09, 0.00)	0.61 (.060)
QA	+0.87 (0.76, 1.11)	+0.27 (0.04, 0.48)	0.90 (<.001)
Code	+0.57 (0.00, 0.93)	+0.25 (−0.15, 0.59)	0.64 (.048)

Table 6: Size effects on T_u, with 95% CIs for the ladder contrasts. The Qwen ladder gains range on every task, though the code CI barely clears zero. The Llama ladder gains on QA only.

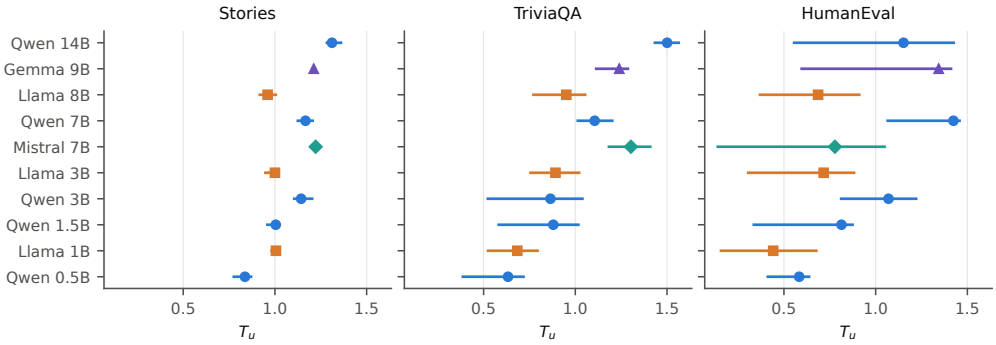


Figure 5: Upper bound of the useful range for each model, ordered by size, with 95% CIs. Story T_u shows Llama holding near 1.0 at every size while Qwen climbs with size. Code CIs are wide, so single-model code comparisons are weak.

Failure modes. Above the cliff, models stop ending their outputs, drift into other languages, and emit code tokens in prose (table 8). Story truncation at the token limit goes from 14% at $T = 1.0$ to 70% at $T = 1.4$.

5 Discussion

What do the cliffs mean? Temperature's useful range is set by where each model breaks, and fixed thresholds such as 0.7, 0.8, or 1.2 describe no single model well. The abrupt shape would fit a feedback process in which one random token that breaks the text makes further random tokens more likely, but we did not test this mechanism. We also cannot say why families differ. Logit scale and vocabulary are candidates, and we measured neither.

What should practitioners do? Below $T \approx 0.8$, temperature was safe on nearly every model we tested, but it is a weak diversity control. The common 0.7 default is therefore safe rather than optimal. Single-attempt code and QA do best at low T. If several attempts can be checked, moderate T helps: pass@4 and coverage@4 both gain over greedy. Above $T \approx 1.0$, the same setting can give intact output on one model and word salad on another of the same size, so the range has to be checked per model. At $T = 1.2$, Mistral-7B has a coherent rate of 0.995 and Llama-3.1-8B has 0.07.

Limitations. Several limits bound these conclusions.

- **One judge with a compressed scale.** The Qwen3-8B judge rarely scores below 3 and shares a lineage with the Qwen subjects. An audit of 16 boundary stories agreed with the threshold on 12 (75%), with the judge slightly conservative. The audit was done by the research agent rather than by human raters, covers one model, and is small.
- **Embedding diversity misleads on degenerate text** (section 4.1).
- **Confounds.** Each family has one checkpoint per size, so family cannot be separated from checkpoint or quantizer differences, and precision co-varies with size. The Llama ladder also crosses versions 3.2 and 3.1.
- **Mistral's low code score.** Its pass@1 of .268 is far below what its QA score suggests. Restoring prompt imports raised it only to .280, and most failures are real assertion errors, but the cause is unexplained. This score should not be taken as representative of Mistral-7B.

Pair (7–9B, all AWQ)	Shape gap	Ts with CI excluding 0	Peak T
Qwen 7B vs. Llama 8B	**0.220**	9	1.4 vs. 1.0
Llama 8B vs. Mistral 7B	0.205	**10**	1.0 vs. 1.4
Llama 8B vs. Gemma 9B	0.118	8	1.0 vs. 1.4
Qwen 7B vs. Gemma 9B	0.104	8	1.4 vs. 1.4
Mistral 7B vs. Gemma 9B	0.098	9	1.4 vs. 1.4
Qwen 7B vs. Mistral 7B	0.046	2	1.4 vs. 1.4
Floor: seed replicate	0.031	0	1.2 vs. 1.2
Floor: bf16 vs. AWQ	0.026	1	1.4 vs. 1.4

Table 7: Family differences in semantic diversity curves, out of the 10 temperatures above 0. Llama pairs with Qwen and Mistral sit far above both floors, and Qwen and Mistral sit close to them. The largest values are in **bold**.

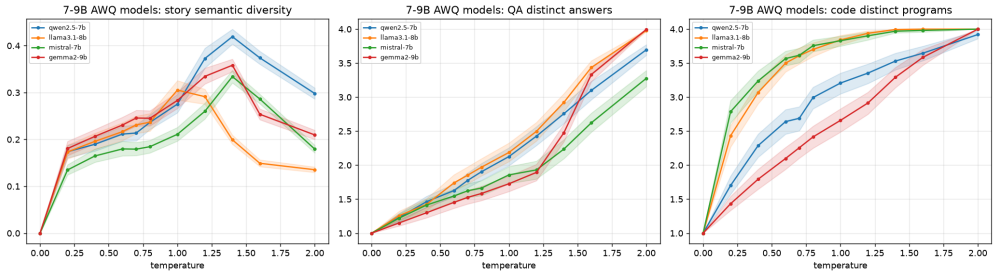


Figure 6: Diversity curves for the four 7–9B AWQ models. On stories, Llama peaks and falls earlier than the others. On code, Llama and Mistral become diverse at low temperature, while Qwen and Gemma rise slowly, so the family order is not the same on each task.

- **Scope.** We test pure temperature only, short stories, one prompt per task, and 4 samples per cell. $T = 0$ uses one sample, one seed replicate on one model is a thin noise floor, and vLLM batch composition can change outputs slightly. Results may not apply to hosted frontier models.
- **Resolution and thresholds.** Grid steps of 0.2 above $T = 0.8$ limit how precisely we can place a cliff, and T_u depends on the 90% threshold. Story T_u CIs are narrow partly because of interpolation.
- **Multiple comparisons** are corrected within tests but not across all hypotheses, and several analysis choices came after seeing partial data (section 3).
- **Contamination.** All three datasets are public and may be in training data. This matters little for curve shapes and more for absolute levels.

Broader implications. Interfaces that offer a single temperature slider imply that a value means the same thing on every model, and our results show it does not. Benchmarks that compare models at one fixed temperature above 1 may partly measure where each model's cliff falls. Diversity metrics also need care, because a metric that falls on broken text can hide what happens past the cliff.

6 Conclusion

We measured diversity and quality on one grid of 10 open-weight models, 11 temperatures, and three tasks. Quality stays nearly flat and then falls at a model-specific point, with story T_u ranging from 0.84 to 1.31. Code has no sharp collapse just above 0.8, factual accuracy loses only a few points above 0.7 with no coverage gain, and the useful range grows with size in Qwen but mostly not in Llama. The main takeaway is that there is no universal temperature: the useful range is a property of the model, not of the setting.

Future work should measure logit scale and top-token margin to test whether they predict the cliff location, add more checkpoints per family to separate family from checkpoint, and run hosted frontier models through the same grid. Sampling at steps of 0.05 around each cliff would test how abrupt

	Code			QA		Stories	
T	Passes	Syntax errors	Truncated	Correct	Rambling	Truncated	Non-ASCII
0.7	52%	2%	2%	49%	3%	10%	0.1%
1.0	48%	4%	2%	47%	2%	14%	0.5%
1.2	42%	17%	9%	43%	4%	47%	1.7%
1.4	32%	45%	35%	37%	12%	70%	5.5%
1.6	21%	68%	61%	29%	28%	87%	9.0%
2.0	5%	93%	89%	11%	58%	92%	11.4%

Table 8: Failure modes pooled over the 10 main models. Rambling means a wrong QA answer of more than 8 words. Between $T = 1.0$ and 1.4, models stop ending their outputs and code stops parsing.

it really is, and a diversity measure that does not collapse on degenerate text, together with human coherence ratings, would remove the two largest measurement limits.

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